

FuDA-YOLO: Low-light Object Detection using Multi-scale Fusion Domain Adaptation

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Abstract—Low-light Object Detection (LOD) is a significant research field in computer vision. In some application fields of automatic driving, security monitoring and so on, it has great value and broad prospect. Low-light conditions means that environment light is insufficient, the loss of image information is serious, and the noise increases. However, most of the traditional object detection methods consider only normal light conditions and their models are also trained that way, leading to a poor performance. A common solution is to use image enhancement technology to improve the brightness of an image and furthest restore its details, and then use a normal object detection network to complete the detection task. These methods cannot well meet the real-time requirements and the evaluation criteria of image enhancement are mostly based on human vision, not suitable for machine vision tasks. Moreover, one of the difficulties of LOD is the lack of annotation data, and it is difficult to collect or make new low-light datasets. To solve these problems, we propose FuDA-YOLO, a LOD method by introducing unsupervised domain adaptation into the YOLO network. The main structure of our FuDA-YOLO is a Multi-scale Fusion Domain Adaptation Module combined with YOLOv8n, to reduce the domain difference between source and target domains. We have carried out a series of experiments to prove that our method can significantly improve model's detection accuracy, domain adaptation ability and inference speed.

Keywords—Low-Light Object Detection, Domain Adaptation, Multi-scale Feature Fusion, Low-Light Image Enhancement

I. INTRODUCTION

Object detection aims to identify and locate several target instances from images or videos, and give bounding boxes and category annotations of the detected objects [1]. In recent years, object detection technology has made significant progress as an important research direction in computer vision, and has played an important role in many application fields such as autonomous driving, security investigation, intelligent video surveillance, and medical image analysis [2]. However, actual application scenarios often require that the object detection model not only meets real-time performance, but also has the adaptability to work under different lighting conditions, perspective changes, and background complexity. For example, object detection in low-light environments still faces challenges with weak ambient light, limited line of sight, and certain dangers. First, in terms of data collection, Low-light Object Detection (LOD) is likely to be applied to special tasks like night security and underground operations, which make it much harder to collect high-quality,

large-scale data. Second, the characteristics of low-light images bring difficulties to manual annotation, such as complex light sources, insufficient brightness, and severe lack of contrast, which also limit the accuracy and robustness of object detection models. Briefly, there is a data scarcity problem of publicly available datasets [3].

To address the above issues, we can use a transfer learning strategy to transfer the knowledge in the well-annotated dataset of normal light to those low-light images. This strategy can be explained by the theory of Domain Adaptation (DA). Specifically, we can regard the rich annotated dataset of normal light as source domain, and the relatively scarce unannotated low-light dataset as target domain. Instead of using the normal light source domain for model training and directly testing on the low-light target domain, it is necessary to reduce the domain differences (such as lighting, contours, etc.) between the two datasets of source and target domains to improve model's abilities of feature extraction and classification.

In this paper, we explore and propose a multi-scale fusion domain adaptive LOD method FuDA-YOLO, which introduces DA into the object detection model YOLOv8 and achieves unsupervised LOD by aligning source and target domains from the perspective of image-level feature alignment. Through a series of experiments, FuDA-YOLO has been proved to bring improvements in both detection accuracy and inference speed, compared to other methods trained only on source domain datasets, and also has better performance in domain adaptation.

II. RELATED WORK

A. Low-light Object Detection

Study via Low-light Image Enhancement. LOD faces a lack of high-quality annotated data, so some studies focus on restoring and enhancing low-light images to solve this problem. Traditional Low-light Image Enhancement (LIE) algorithms improve the dynamic range and contrast of images by adjusting the detailed information of them [4]. After the rise of deep learning theory, many researchers have applied it to LIE, including supervised methods [5] and unsupervised ones [6], and some zero-shot learning methods [7, 8]. In addition, some studies adding image processing modules to object detection networks to enhance and train images [9, 10]. This kind of study undoubtedly increases the reasoning time and meets the high real-time requirements in real-world scenarios.

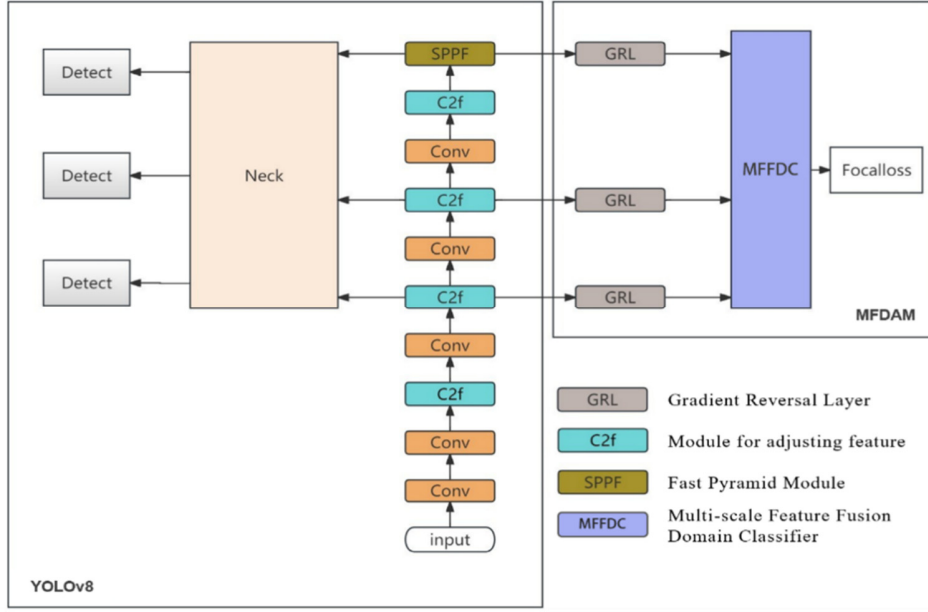


Figure 1. Overall structure of FuDA-YOLO, which is composed of the YOLOv8 backbone and our module MFDAM. Among them, We introduce sub modules like GRL, C2f, and MFFDC, to achieve our goal of multi-scale fusion. For model training, three feature layers in YOLOv8, which are input to the Neck, are used as the input of the domain classifier, and they are gradually fused into one layer through convolution, downsampling, and Concat operations, with the help of the C2f module. After that, CBAM are introduced.

Direct Study on Low-light Object Detection. Some methods try not to rely on LIE algorithms, but directly design and optimize models based on the characteristics of low-light images [11, 12]. These studies train end-to-end models but are essentially supervised object detection methods that assume the existence of high-quality annotated images for training. In practical applications, it is costly to collect abundant low-light images and annotate them.

B. Domain Adaptive Object Detection

DA aims at bridging the distribution difference between source and target domains so that the model trained in source domain can perform well in target domain. Study on DA can be roughly divided into three categories: difference-based methods, adversarial methods, and special ones such as reconstruction-based ones [13].

Compared with DA, Domain Adaptive Object Detection (DAOD) has clear downstream goals. Methods of DAOD are often improved from existing object detection methods, which can be classified into self-training-based, teacher-student model-based, and adversarial methods. DAOD based on self-training is to use a self-training strategy to generate pseudo labels in target domain to retrain the detector [14]. This kind of methods strictly demand of a small domain difference between source and target domains to ensure pseudo-label quality. Besides, some studies use Teacher-student Model [15, 16], which is a common technique in deep learning used to transfer the knowledge of a complex model to a student model. Adversarial methods of DAOD derives from the adversarial DA methods mentioned above. Chen et al. proposed the first method to apply DA to object detection [17]. In subsequent studies, researchers are committed to designing an effective discriminator for detection in different feature spaces such as image level [18], pixel level [19], and instance level [20].

III. METHODOLOGY

We propose FuDA-YOLO, which introduces a key module called Multi-scale Fusion Domain Adaptation Module (MFDAM) into the YOLOv8 network using labeled source domain datasets of normal light and trained on unlabeled target domain datasets of low light, can not only improve the accuracy of LOD, but maintain the high-speed advantages of the YOLO model itself, making it possible for real-time applications. The overall structure of our FuDA-YOLO is shown in Figure 1. In this chapter, we first review YOLOv8 in Section A, which constitutes the main network structure and baselines in our work. Section B presents the overall design and detailed structure of MFDAM. In Section C we introduce our loss function for the domain classifier.

A. YOLOv8

Since the YOLO detection network was published in 2016, its versions has been continuously updated and iterated for better performance. Among them, YOLOv8 [21] is a newer version launched by the Ultralytics LLC team in 2023. It builds on the success of YOLOv5 and adds new methods to improve model robustness and flexibility. YOLOv8 supports multiple tasks like image classification, object detection, and instance segmentation. Its main structure consists of Input, Backbone, Neck, Head, and the loss function. An input image is scaled to a fixed size (640*640) through Input and enhanced. Backbone downsamples the feature to its 1/32. Neck is used to fuse and enhance the multi-scale features from Backbone. Head regards classification and bounding-box regression as two independent tasks, and adopts Task-Aligned Assigner [22] as the distribution strategy for positive and negative samples. Finally, the loss function includes two branches Cls and BoxReg. Cls uses VFL loss [23] and BoxReg consists of CioU [25] and DFL [26].

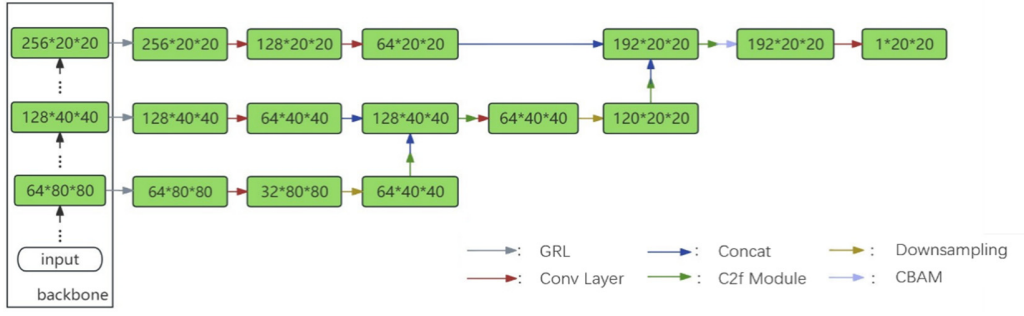


Figure 2. Detailed structure of our MFFDC based on YOLOv8n. MFFDC can comprehensively understand and classify multi-scale features, and complete the fusion of input features of different scales through operations such as convolution, Concat, and downsampling, and then perform classification, thus balancing semantic information expression and resolution.

B. Multi-scale Fusion Domain Adaptation Module

Our MFDAM consists of several key components: Gradient Reversal Layer (GRL), which reverses the gradients during the gradient descent operation to confuse source domain with target domain; Multi-scale Feature Fusion Domain Classifier (MFFDC), which uses convolution and Concat to fuse features of different scales for domain classification; the C2f module, which gradually adjusts and enriches the fused features through a series of convolutional layers with residual connections; Convolutional Block Attention Module (CBAM) [27], which recalibrates features across channels and spatial locations to highlight low-light information areas and suppress noises.

Figure 1 shows the locations of MFDAM and YOLOv8 in our network. Actually we take YOLOv8n as the baseline for our experiments. During the prediction process, only low-light images are input, meanwhile the DA module will not participate in reasoning. Only a YOLOv8n structure is involved, so our method will not bring extra reasoning time. Below, we will introduce each component in MFDAM.

1) *Gradient Reversal Layer*: YOLOv8 hopes to reduce the domain difference during its feature extraction while ensuring high detection performance, nevertheless, the purpose of a domain classifier is the opposite. It tries to distinguish whether the feature comes from source or target domain as much as possible by minimizing a binary classification loss. To achieve these two opposite goals, GRL needs to be introduced. GRL is generally used to reverse the gradient of a specific layer during training. In the forward propagation, GRL itself does not change any input data with an identity operation. During backpropagation, GRL multiplies the input gradient by a negative constant λ , thereby reversing its direction, as shown in (1). This adversarial method forces the model to learn features that cannot be easily distinguished from its domain while trying to optimize the object detection task, thereby enhancing model's generalization ability to new domains. Briefly, adding GRL between input features and the domain classifier in our MFDAM can achieve the goal of DA.

$$\frac{\partial L}{\partial x} = -\lambda \frac{\partial L}{\partial y} \quad (1)$$

2) *Multi-scale Feature Fusion Domain Classifier*: As mentioned above, YOLOv8 sends multi-scale input features

into the neck structure and generates three sets of features of different scales, which are processed by GRL and output to MFFDC. To minimize the domain difference between source and target domains, the domain classifier of our FuDA-YOLO should receive multi-scale features as its input. However, conducting convolution and classification separately of three feature sets using multiple independent domain classifiers will lose feature information at different scales, making it impossible for the domain classifier to fully characterize the input data. The resolution of shallow network features is high and the receptive field is small, but with a weak semantic representation, meanwhile the semantic representation of deep network features is strong, but with a low resolution. Thus, we use a multi-scale feature fusion method to fuse the features of different scales, which are then classified. Its specific structure is shown in Figure 2.

MFFDC accepts features of three different scales and sizes as input. For lowest level features, their channel number is halved through a 1×1 convolution kernel. Then, they are downsampled to reduce the spatial dimension and expand the receptive field. They are then concatenated with middle-level features that have similarly had their channel number reduced through a 1×1 convolution kernel. The features after Concat are then halved and downsampled again and then concatenated with highest-level features that have undergone twice channel halving operations. Finally, they are concatenated with a 1×1 convolution kernel to reduce the channel number to 1 to obtain the final output. During the fusion process, the features are also processed by the C2f module and CBAM, which will be described in detail below.

3) *C2f Module for Feature Fusion*: Relying solely on operations such as convolution, downsampling, and Concat in the feature fusion network will lead to information redundancy or loss of different scales. In addition, the feature fusion network uses deeper features for domain classification, which has a certain risk of gradient vanishing. Therefore, we introduce a C2f module to improve the domain classifier, as shown in the upper part of Figure 3. For a feature with C input channels, the C2f module performs the following operations: After being processed by a 1×1 convolutional layer, the feature is split into two features with $C/2$ channels, one of which is processed by several Bottleneck layers (the number of Bottleneck layers is

controlled by a parameter n), and then concatenated with another. To get the final result, it is then restored through another 1×1 convolutional layer. The Bottleneck layer is a standard bottleneck residual structure, shown in the lower part of Figure 3. In Bottleneck, the input feature firstly passes through the first convolutional layer to halve its channel number, then passes through the second one to restore its original channel number. Finally, the original input feature and the processed feature get a residual connection.

4) *Convolutional Block Attention Module*: To reduce the impact of noise on domain classification, to guide the model to focus on the channel that best reflects the light, and pay more attention to those image areas which have the largest light difference between source and target domains, we introduce CBAM [27] in MFDAM. As shown in Figure 4, CBAM uses Channel Attention (CA) and Spatial Attention (SA) to focus on important features. CA determines the importance of each channel by global information. The input features processed by CA are then input into the SA module, which helps the model focus on spatial locations with more information. CBAM can effectively enhance model's ability to perceive key features, making it perform better in domain classification.

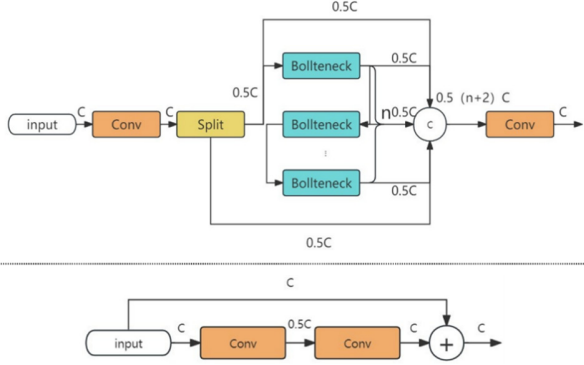


Figure 3. Structure of the C2f module in the upper part and its Bottleneck layer in the lower part. C2f module can enhance the ability of our model to integrate features through Convolution, Split, and Bottleneck layer, thereby improving object detection performance. Bottleneck is introduced to reduce the risk of gradient vanishing.

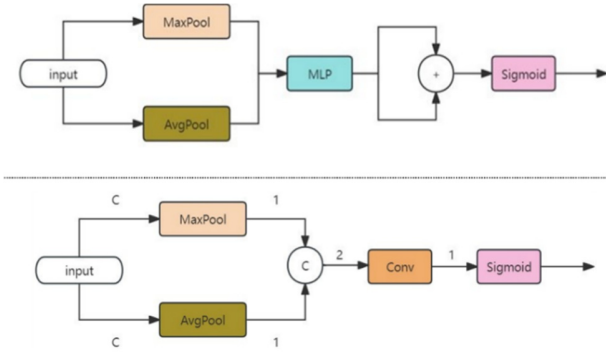


Figure 4. Main content of Convolutional Block Attention Module (CBAM), used to reduce the impact of noise in low-light images on domain classification performance. CA and SA together constitute CBAM. The former is used to emphasize those channel features that are more significant for domain classification tasks, and the latter is for capturing more spatial information.

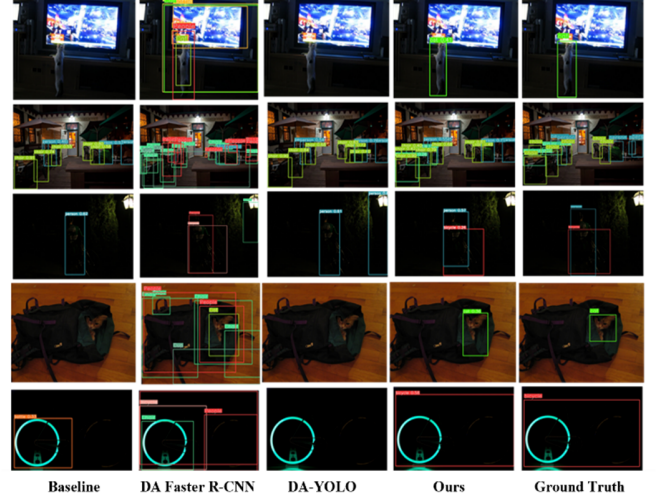


Figure 5. Several detection results on ExDark dataset from the baseline, other models and ours. Compared with the detection results of DA-YOLO and DA-Faster R-CNN which also have good performance, our model has better accuracy and less missed detection in the face of uneven light, low-light and unclear objects.

C. Loss function for MFFDC

The domain classifier tries to distinguish whether a feature comes from source or target domain by minimizing the binary classification loss, which is traditionally a binary cross-entropy loss function (BCE). However, in a low-light image domain classification problem, BCE cannot pay enough attention to a small number of samples that are difficult to distinguish and thus, make them overwhelmed by massive easy-to-classify samples during the training process. Therefore, we use the Focalloss [24] function instead shown in (2). In this equation, p_t is the predicted probability of the network output. γ is an adjustable value called the focus parameter, used to control the weight of easy-to-classify samples. The larger γ is set, the more the model focuses on the hard-to-classify samples in the data set. When a sample is correctly classified, p_t tends to be 1, and the loss is close to 0; when a sample is misclassified or hard to classify, p_t will be small and the loss gets larger, but due to the existence of $(1-p_t)^\gamma$, the loss of easy-to-classify samples will be multiplied by a coefficient very close to 0, which reduces the weight of this part of total loss.

$$FL(p_t) = -1/N \sum_{t=1}^N (1-p_t)^\gamma \cdot \log(p_t) \quad (2)$$

IV. EXPERIMENT

In this chapter, we introduce the datasets used in our experiment in Section A, the details of our experiment in Section B, and then the main results of some comparative and ablation experiments in Section C and D respectively.

A. Implementation Details

In our model training, PASCAL VOC (VOC) dataset is used as source domain dataset and ExDark dataset is used as target domain dataset. VOC here is a mix of VOC 2007 and 2012. To align the category samples between two datasets, the images are screened before training. There are 10 same categories in the two datasets, namely *bicycle*, *boat*, *cup*, *bus*, *car*, *cat*, *chair*, *dog*,

motorcycle, and *person*, and delete the rest together with those images that contain no category after processing. We also retains a pair of categories with certain similarities in features: *dining table* (VOC) and *table* (ExDark), to verify that our DA method is still robust when category features have some domain differences. After screening, the annotated VOC training set in source domain contains 12,540 images, and the unannotated ExDark dataset contains 2,870 images.

For hardware and software configurations, we use GeForce RTX 3090 GPU, Ubuntu 20.04.6 as OS, CUDA 12.1, torch 2.1.0 as deep learning framework and Python 3.9.18 for training. As to the basic object detection network, we use the lightest version YOLOv8n, with the depth factor set to 0.33 and the width factor set to 0.25. For hyperparameter, we select the SGD optimizer with a momentum coefficient of 0.9, 200 epochs, and a batch size of 32, respectively with 16 batches from the dataset of source and target domains. The value of γ in Focalloss is set to 2, and the parameter n in Bottleneck submodules of our C2f module is 2, 1, 2, 1.

B. Main Results

Comparison with other Domain Adaptation Models. We train the YOLOv8n model on source domain dataset and test it on ExDark dataset as Baseline mentioned below. We select DA-Faster R-CNN [17], DA-YOLO [28] reproduced by us on YOLOv8 to eliminate the impact of YOLO versions, a LOD model via image enhancement (IAT) [29], and an unsupervised Teacher-student model-based object detection model 2PCNet [30] for our experiment. Each of them uses the VOC dataset for source domain and ExDark for target domain. Table 1 shows the results of our experiment, where DA-F and DA-Y refers to DA-Faster R-CNN and DA-YOLO. Our model achieves 54.3% mAP, 2.5% higher than Baseline and 0.7% higher than the second-best 2PCNet.

When compared with Baseline, ours performs better for most categories except Bottle which has obscure object outlines and features. Cat and Dog also meet lower mAP because their features are highly similar and mostly in a small region than DA-Faster R-CNN, which was specially designed for feature-level domain alignment. This could be a improvement direction in our future work. Figure 5 shows several detection results from all above models on ExDark dataset.

Comparison of Inference Speed with other Domain Adaptation Models. We use Frames Per Second (FPS) as an indicator to evaluate the real-time performance of those models, and taking averagely ten tests for each model as the final result, as shown in Table 2. DA-YOLO and ours do not introduce extra calculations during inference, but only use the original YOLOv8 network for inference, so their FPS is the best and highly similar to YOLOv8n; IAT enhances image features and then makes inference as a essentially two-stage method, so it has a lower FPS; DA-Faster R-CNN and 2PCNet have the lowest FPS based on the two-stage model FasterR-CNN [31] for their inference.

C. Ablation Study

Table 3 shows the mAP of each module before and after ablation. In Baseline, only MFFDC is added. As shown in the table, the C2f module brings a significant performance

improvement, and mAP will no longer significantly improves when the number set of Bottleneck in C2f reaches 2 after both downsampling and Concat, therefore, we selected 2, 1, 2, 1 for the n Bottlenecks in C2f. When only CBAM is added based on MFFDC, mAP has a increase. Combining the advantages of CBAM in classification and of C2f in fusion, our model achieves a further improvement, as shown, mAP50 increases from 52.1% to 53.4% and mAP50:95 from 29.3% to 30.5%.

Table 1. Detection accuracy of several models on ExDark.

Class	AP (%)					
	Baseline	DA-F	DA-Y	IAT	2PCNet	Ours
bicycle	62.8	69.8	65.1	63.8	69.7	69.9
boat	36.4	30.1	36.2	36.6	28.6	37.2
bottle	42.5	38.6	40.5	38.9	44.4	41.9
bus	74.1	75.9	74.3	70.0	72.6	75.2
car	61.5	61.2	59.9	55.1	62.3	62.4
cat	56.0	63.9	58.7	50.4	59.1	61.9
chair	45.4	44.5	45.6	41.4	44.7	47.1
(dining) table	25.0	21.4	25.0	23.3	25.2	25.4
dog	52.3	64.8	53.6	50.5	62.5	54.3
motor-bike	50.5	51.6	53.1	48.7	55.0	55.6
person	63.6	61.7	64.4	59.2	65.4	66.3
map	51.8	53.0	52.4	48.9	53.6	54.3

Table 2. Comparison of model inference speed.

Model to make inference	FPS (f/s)
Baseline	251.3
DA Faster R-CNN	21.5
DA YOLO	245.6
IAT	72.4
2PCNet	20.0
Ours	244.7

Table 3. Model performance with different modules.

Baseline	C2f	CBAM	mAP0.5 (%)	mAP0.5:0.95 (%)
√			52.1	29.3
√	√		53.1	30.1
√		√	52.8	29.9
√			52.3	29.5
√	√	√	53.4	30.5
√	√	√	54.3	31.0

V. CONCLUSION

In this paper, we propose FuDA-YOLO, a low-light object detection method using unsupervised Domain Adaptation (DA). Its core component is a multi-scale fusion domain adaptation module (MFDAM) based on YOLOv8 network. It successfully introduces domain adaptation technology into YOLO network to improve the scarcity of low-light annotated data. MFDAM reduces the domain difference between source and target domains at multiple scales based on a Gradient Reversal Layer and a Multi-scale Feature Fusion Domain Classifier. It also uses the C2f module, CBAM, and Focalloss to further improve model's detection accuracy and inference speed.

In our future work, the YOLOv8 network structure can be further modified to improve its detection ability in low-light scenes, based on our MFDAM. In addition to those pixel-level and image-level DA methods mentioned in this paper, instance-level methods can be also introduced to solve object detection and recognition tasks with those highly similar categories.

REFERENCES

- [1] Zaidi, S.S.A.; Ansari, M.S.; Aslam, A.; Kanwal, N.; Asghar, M.; Lee, B. A survey of modern deep learning based object detection models. *Digit. Signal Process. A Rev. J.* 2022, 126, 1–18.
- [2] Zou, Z.; Chen, K.; Shi, Z.; Guo, Y.; Ye, J. Object detection in 20 years: a survey. *Proc. IEEE* 2023, 111, 257–276.
- [3] Li, C.; Guo, C.; Han, L.; Jiang, J.; Cheng, M.M.; Gu, J.; Loy, C.C. Low-Light image and video enhancement using deep learning: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44(12), 9396–9416 (2021).
- [4] Guo, X.; Li, Y.; Ling, H. LIME: Low-light image enhancement via illumination map estimation. *IEEE Trans. Image Process.* 2016, 26, 982–993.
- [5] Feng, Yixu et al. You Only Need One Color Space: An efficient network for low-light image enhancement. *ArXiv abs/2402.05809* (2024): n. pag.
- [6] Jiang Y, Gong X, Liu D, et al. Enlightengan: Deep light enhancement without paired supervision. *IEEE transactions on image processing.* 2021, 30: 2340-2349.
- [7] Zhang L, Zhang L, Liu X, et al. Zero-shot restoration of back-lit images using deep internal learning. *Proceedings of the 27th ACM International Conference on Multimedia.* 2019: 1623-1631.
- [8] Guo C, Li C, Guo J, et al. Zero-reference deep curve estimation for low-light image enhancement. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition.* 2020: 1780-1789.
- [9] Kalwar S, Patel D, Aanegola A, et al. Gdip: Gated differentiable image processing for object detection in adverse conditions. *2023 IEEE International Conference on Robotics and Automation (ICRA).* 2023: 7083-7089.
- [10] Yin X, Yu Z, Fei Z, et al. Pe-yolo: Pyramid enhancement network for dark object detection. *International Conference on Artificial Neural Networks.* 2023: 163-174.
- [11] Wu S, Liu Z, Lu H, et al. Shadow Hunter: Low-Illumination Object-Detection Algorithm. *Applied Sciences*, 2023, 13(16): 9261.
- [12] Hong, Mingbo et al. You Only Look Around: Learning illumination invariant feature for low-light object detection. *ArXiv abs/2410.18398* (2024): n. pag.
- [13] Mohamadi, H., Keyvanrad, M.A., & Mohammadi, M.R. (2024). Feature Based Methods in Domain Adaptation for Object Detection: A Review Paper.
- [14] Kennerley M, Wang J G, Veeravalli B, et al. 2pcnet: Two-phase consistency training for day-to-night unsupervised domain adaptive object detection. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition.* 2023: 11484-11493.
- [15] Cao S, Joshi D, Gui L Y, et al. Contrastive mean teacher for domain adaptive object detectors. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition.* 2023: 23839-23848.
- [16] Zhao Z, Wei S, Chen Q, et al. Masked retraining teacher-student framework for domain adaptive object detection. *Proceedings of the IEEE/CVF International Conference on Computer Vision.* 2023: 19039-19049.
- [17] Chen Y, Li W, Sakaridis C, et al. Domain adaptive faster r-cnn for object detection in the wild. *Proceedings of the IEEE conference on computer vision and pattern recognition.* 2018: 3339-3348.
- [18] Li C, Du D, Zhang L, et al. Spatial attention pyramid network for unsupervised domain adaptation. *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XIII* 16. 2020: 481-497.
- [19] Hsu H K, Yao C H, Tsai Y H, et al. Progressive domain adaptation for object detection. *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision.* 2020: 749-757.
- [20] Jiang J, Chen B, Wang J, et al. Decoupled adaptation for cross-domain object detection. *arXiv preprint arXiv:2110.02578*, 2021.
- [21] Varghese, Rejin and Sambath. M. YOLOv8: a novel object detection algorithm with enhanced performance and robustness. *2024 International Conference on Advances in Data Engineering and Intelligent Computing Systems (ADICS)* (2024): 1-6.
- [22] Feng C, Zhong Y, Gao Y, et al. Tood: Task-aligned one-stage object detection. *2021 IEEE/CVF International Conference on Computer Vision (ICCV).* 2021: 3490-3499.
- [23] Zhang H, Wang Y, Dayoub F, et al. Varifocalnet: An iou-aware dense object detector. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition.* 2021: 8514-8523.
- [24] Lin T Y, Goyal P, Girshick R, et al. Focal loss for dense object detection. *Proceedings of the IEEE International Conference on Computer Vision.* 2017: 2980-2988.
- [25] Zheng Z, Wang P, Liu W, et al. Distance-IoU loss: Faster and better learning for bounding box regression. *Proceedings of the AAAI Conference on Artificial Intelligence: vol. 34: 07.* 2020: 12993-13000.
- [26] Li X, Wang W, Wu L, et al. Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection. *Advances in Neural Information Processing Systems.* 2020, 33: 21002-21012.
- [27] Woo S, Park J, Lee J Y, et al. Cbam: Convolutional block attention module. *Proceedings of the European Conference on Computer Vision (ECCV).* 2018: 3-19.
- [28] Hnawa M, Radha H. Multiscale domain adaptive yolo for cross-domain object detection. *2021 IEEE International Conference on Image Processing (ICIP).* 2021: 3323-3327.
- [29] Cui Z, Li K, Gu L, et al. You only need 90k parameters to adapt light: a light weight transformer for image enhancement and exposure correction. *arXiv preprint arXiv:2205.14871*, 2022.
- [30] Lore K G, Akintayo A, Sarkar S. LLNet: A deep autoencoder approach to natural low-light image enhancement. *Pattern Recognition*, 2017, 61: 650-662.
- [31] Ren S, He K, Girshick R, et al. Faster r-cnn: Towards real-time object detection with region proposal networks. *Advances in Neural Information Processing Systems.* 2015, 28.